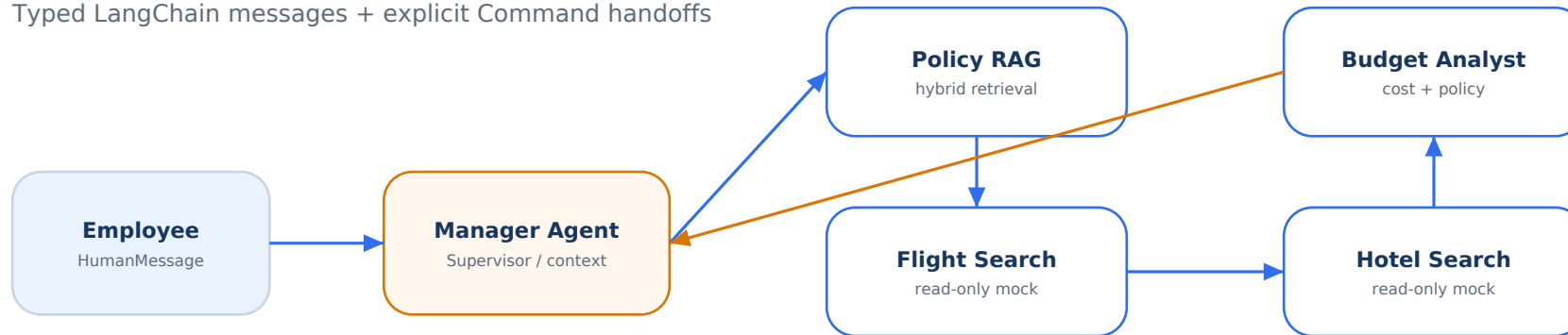


DZ 07 - Supervisor Multi-Agent Architecture

Typed LangChain messages + explicit Command handoffs



TravelState(MessagesState)

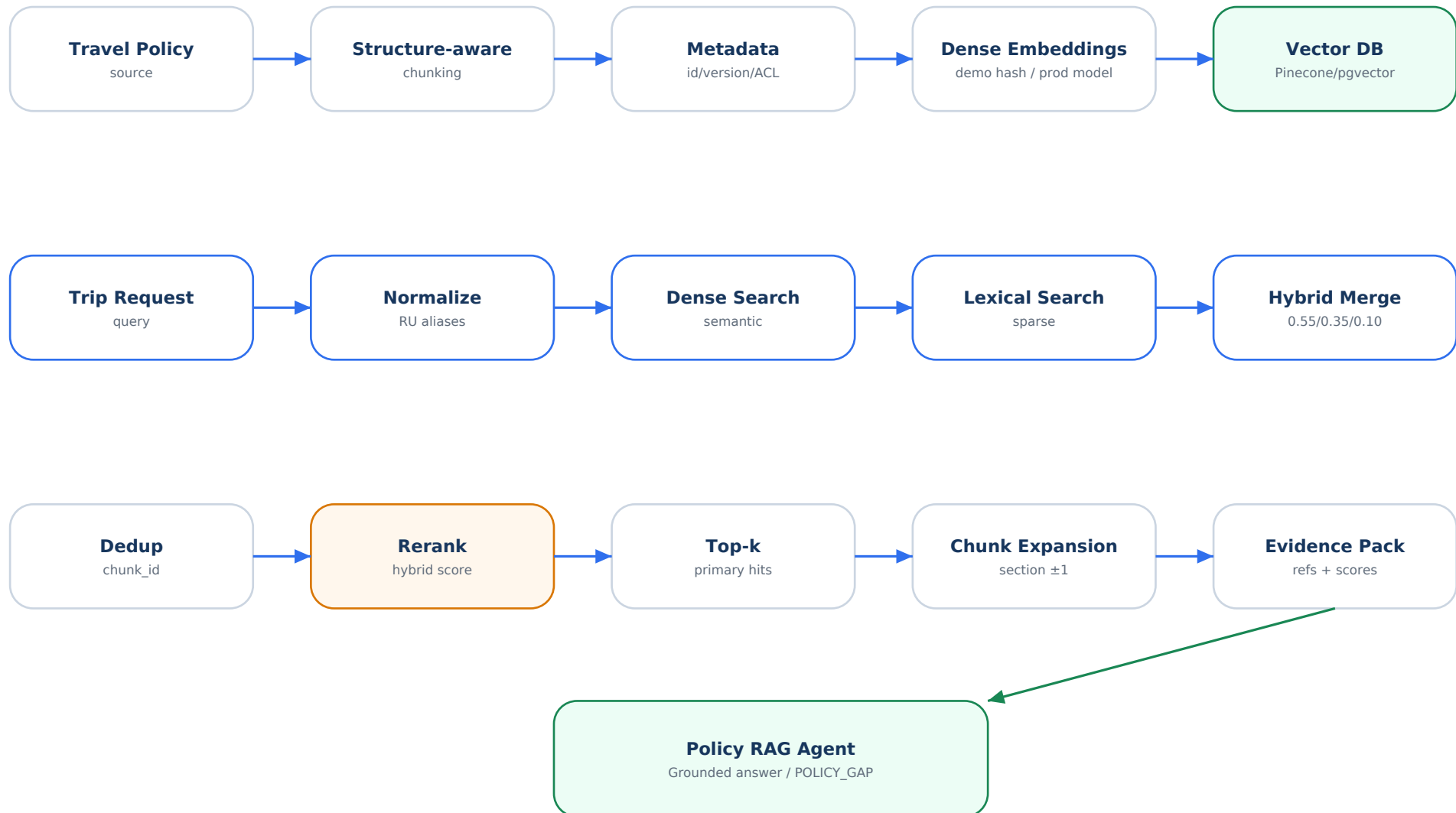
messages: SystemMessage / HumanMessage / AIMessage
request · policy_hits · expanded_policy_chunks
flight_options · hotel_options · budget · final_answer
trace: every explicit handoff

Agent communication gate

- ✓ typed messages are stored in shared state
- ✓ Manager delegates through Command(goto=...)
- ✓ each agent has Single Responsibility
- ✓ final response returns to Manager

Hybrid RAG Flow - Corporate Travel Policy

Dense + lexical retrieval, dedup, reranking, chunk expansion and evidence refs



Normalization: отель→гостиница · Москве→москва · перелета→перелёт · согласования→согласование

Verification & Production Boundary

What the demo proves, and what remains a production replacement

Layer	Demo	Production
Embeddings	deterministic hash	dedicated embedding model
Vector store	in-memory chunks	Pinecone / pgvector
Lexical	token overlap	sparse/BM25-like index
Reranker	hybrid score	cross-encoder / managed
Messages	LangChain typed	typed + persistence/observability
Handoff	Command(goto)	policy-aware supervisor
Eval	5 synthetic cases	versioned representative set

CI quality gate

pytest: typed messages + explicit handoffs
hybrid retrieval: dense + lexical + rerank + expansion
HitRate@2 $\geq$ 0.80
MRR@2 $\geq$ 0.70
demo run: no API key / no network

Final response contract

status + estimated_total_rub
flight_id + hotel_id
evidence_refs with dense/lexical/hybrid scores
retrieval_metrics
expanded_context_chunk_ids